UNIVERSITY OF WOLVERHAMPTON

Collaborative Development 5CS024

Tutorial Week - 1

Collaborative Curriculum Development Team

Welcome to the Collaboration

This module is a bit different than other module. Here, we will be collaborating together and learning the aspect of collaboration and directing the project into a success!!

Collaborative Development Team

Agenda

- Introduction to collaborative development
- Project and Project Management
- Software Development Life-Cycle
- Waterfall Model & Incremental Model

Introduction to Module

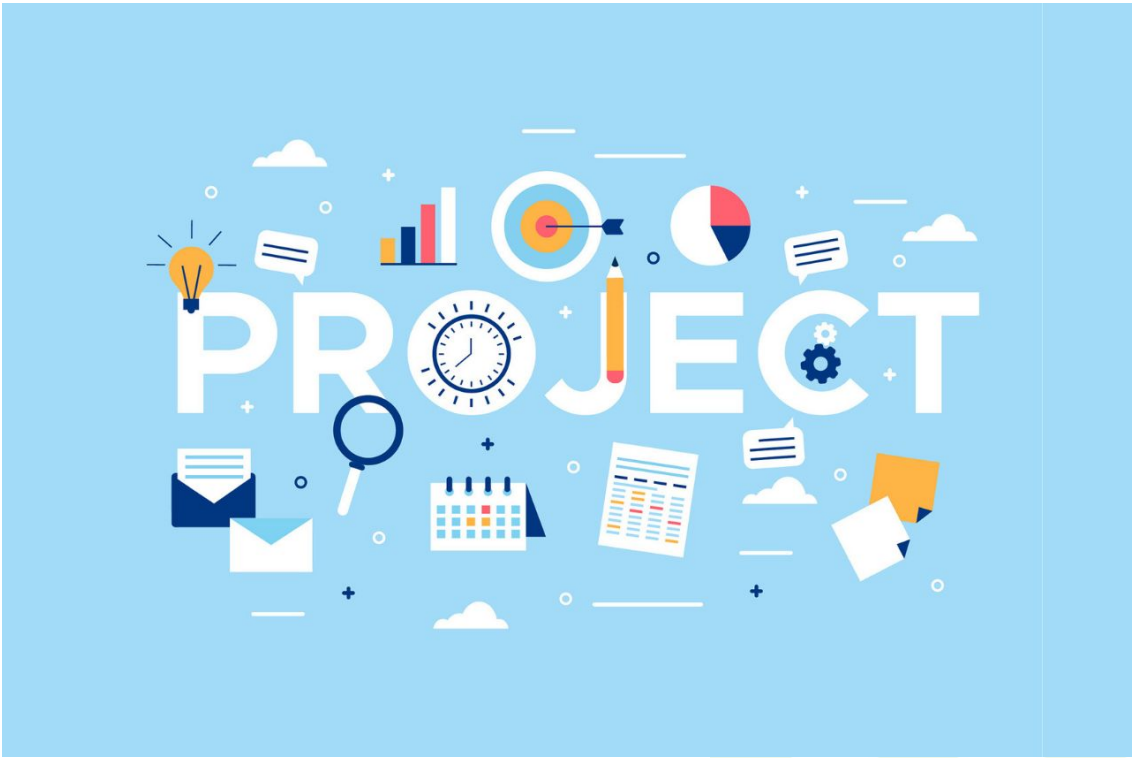
- Collaborative development refers to the process where multiple individuals or teams work together to achieve a common goal. It emphasizes teamwork, communication, and shared responsibility.
- As mentioned earlier, in this module we will be working on Improved Problem – Solving, Efficiency, Innovations and Team Morale.
- We will moving with to the same direction in a team to achieve the project that identify as a problem statement in the most agile, cross-functional and collaborative way.

Since, we will be working on a project so first let’s understand what a project is?

Project

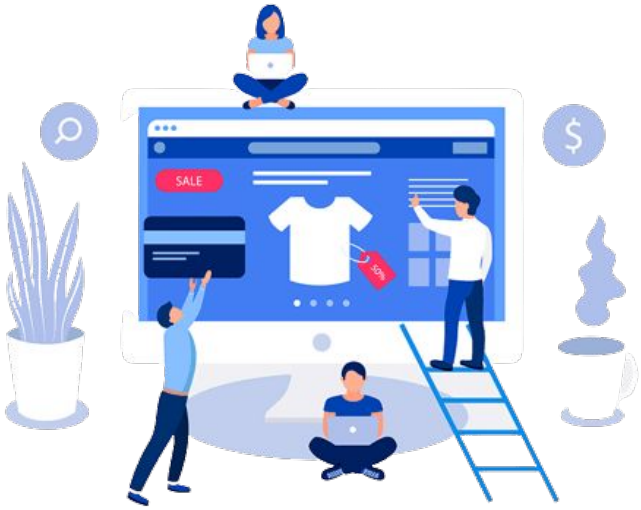
What is a Project?

- A project is "**a temporary endeavor undertaken to create a unique product, service or results**". PMBOK® Guide, Fifth Edition.
- **Temporary:** Defined beginning and end in time.
- **Unique:** Not a routine operation.



Project Cont..

- Project Could be **Software** and **Non-Software**.



Various software examples could be **ecommerce website**, **CRM software**, **Credit card fraud detection.**

Various non – software examples could **be constructing a house, A wedding or even a political election campaign.**

Project Cont..

- Regardless of whether a project is software – based or non- software, it consists of several key aspects:

Project Scope

Project Objectives

Task and Activities

Project Stakeholders

Project Timeline

Resources

Risk Management

Project Success and Failure

Deliverables

Software Project

- For this module, we will be working on software project.
- Software project is a structured effort to develop, maintain or enhance a software system within a defined timeline, budget, and scope. It involves planning, designing, coding, testing and deploying software to meet specific user or business requirement.
- Generally there two **types** of software projects.

Products (Generic Software)

Services (Custom Software)

Types of Software Project

1. **Product Software:** These are those type of software that can be bought easily from the market such as antivirus package, MS word and so on.
2. **Service Software:** These are those type of software that are customized based on the client's requirement.

While managing a software the project managers should be able to manage both type of projects!!

And it is essential for a stakeholders to get the project success!!

Project Management

- Project Management, is the application of knowledge, skills, tools, and techniques to a project activities to meet the project requirements.
- Every projects are managed with the certain steps of Planning, Implementing, Monitoring & Controlling and Closing.
- In **Project Management,** there are multiple frameworks used to plan, execute, and control the projects.

Traditional (Predictive Frameworks)

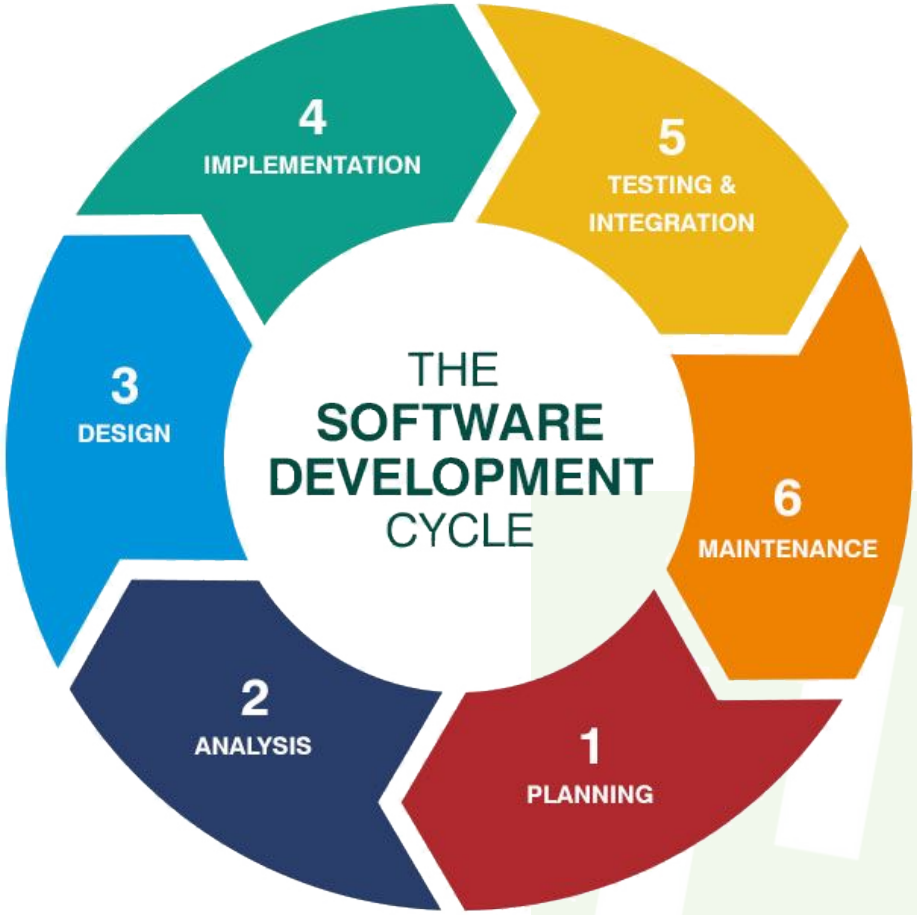
Agile (Adaptive Frameworks)

Hybrid Frameworks

From, a long time software project used to follow the framework build and fix. Later on while working on a team it was essential to follow a framework for better tracking progress

Software Development Lifecycle

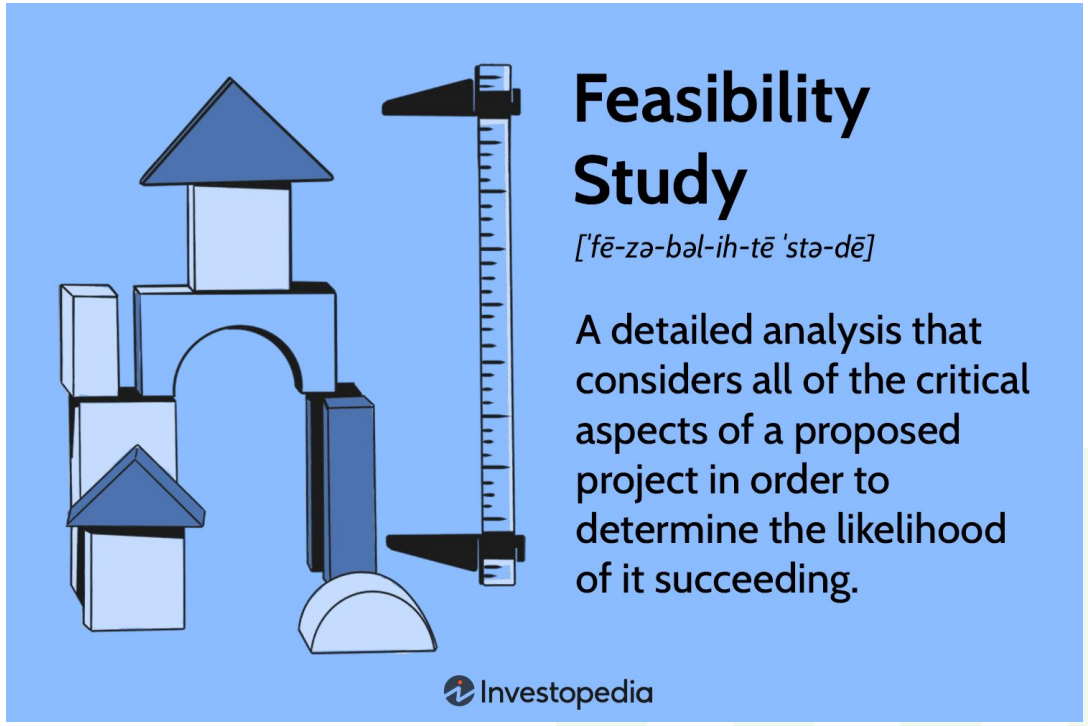
- Process that covers the entire spectrum of software development, from the product's initial conception to its eventual retirement.
- These are derived from various frameworks that we discussed earlier such as PMI (PMBOK) or Prince2
- Further these SDLC are derived into various model such as linear process model and iterative process model



The First Step in any Project Development

- Feasibility Study: 3 types

- **Technical Feasible Study:** The first one is technical feasible study whether the development team has the competence and capability to undertake the project or not
- **Economic Feasibility Study:** Here, need to see either the budget that would be profitable to carry out the project or not
- **Schedule Feasibility Study:** Here , the time that the project managers estimates for the project to complete whether the customer agrees to that time or not.



The main aim of this step is to determine the financially worthwhile and technically feasible

First Case Study

Project Overview: Fit Track: A Fitness Tracking App

Objective:
To develop a mobile app that tracks user's daily activities (steps, calories burned, heart rate) and provides personalized fitness recommendation

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Task for Students
Based on the scenario, answer the following questions:
Technical Feasibility: Does the team have the competence and capability to undertake the project?
Economic Feasibility: Is the budget sufficient, and will the project be profitable?
Schedule Feasibility: Can the project be completed within the estimated timeline?
Overall Feasibility: Should the stakeholders proceed with the project? Why or why not?

First Case Study

Team Details (Technical Feasibility)

Development Team:

- **Front-End Developers**: 2 (experienced in React Native).
- **Back-End Developers**: 2 (experienced in Node.js and MongoDB).
- **UI/UX Designers**: 1 (experienced in mobile app design).
- **QA Testers**: 1 (experienced in mobile app testing).

Team Competence:

The team has previously developed a weather app but has no prior experience in fitness tracking or health-related apps.

They are unfamiliar with integrating wearable devices (e.g., Fitbit, Apple Watch) into the app.

Technology Stack:

- **Front-End**: React Native.
- **Back-End**: Node.js, MongoDB.
- **APIs**: Will need to integrate third-party APIs for wearable devices.

Does the team have the competence and capability to undertake the project?

First Case Study

Economics (Economic Feasibility)

Estimated Costs:

- **Development Costs**: $150,000 (team salaries, software licenses, API subscriptions).
- **Marketing Costs**: $30,000 (social media ads, influencer partnerships).
- **Operational Costs**: $20,000 (server hosting, maintenance, customer support).

Revenue Model:

The app will be free to download but will offer a premium subscription at $5/month.

Expected user base:

- **Year 1**: 50,000 users (10% conversion to premium).
- **Year 2**: 100,000 users (15% conversion to premium).

Year 1 Revenue: $300,000.

Year 2 Revenue: $900,000.

Break-Even Analysis:

The project will break even in **Year 1** if the user base reaches 50,000 and 10% convert to premium.

Is the budget sufficient, and will the project be profitable?

First Case Study

Schedule (Schedule Feasibility)

Estimated Timeline: 9 months

- **Months 1-2**: Requirement gathering and system design.
- **Months 3-5**: Development of the app (front-end and back-end).
- **Months 6-7**: Integration with wearable devices and third-party APIs.
- **Months 8-9**: Testing, bug fixes, and launch preparation.

Customer Expectations:

- The stakeholders want the app to be launched within 9 months.
- They expect a working prototype after 5 months to demonstrate progress.

Risks to Timeline:

- Integrating wearable devices and third-party APIs could take longer than expected.
- The team's lack of experience in fitness tracking apps could lead to delays.

Can the project be completed within the estimated timeline?

Project Development Model

- So far, many life cycles models have been proposed such as

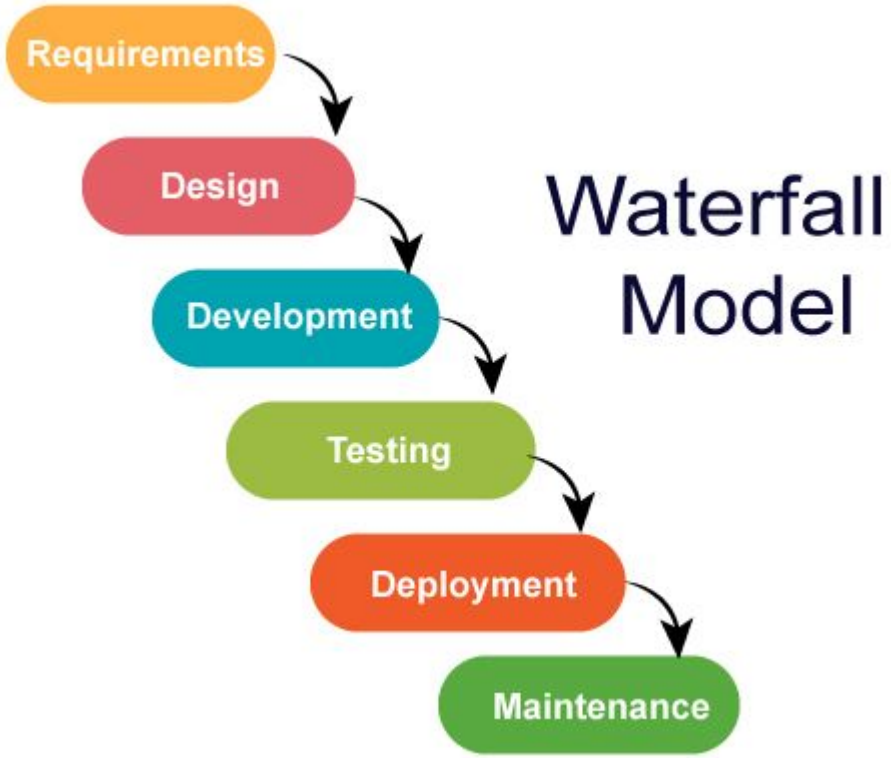
1. Waterfall Model
2. V Model
3. Evolutionary Model
4. Prototyping Model
5. Spiral Model
6. **Incremental Model**
7. **Agile Model**

Traditional Models

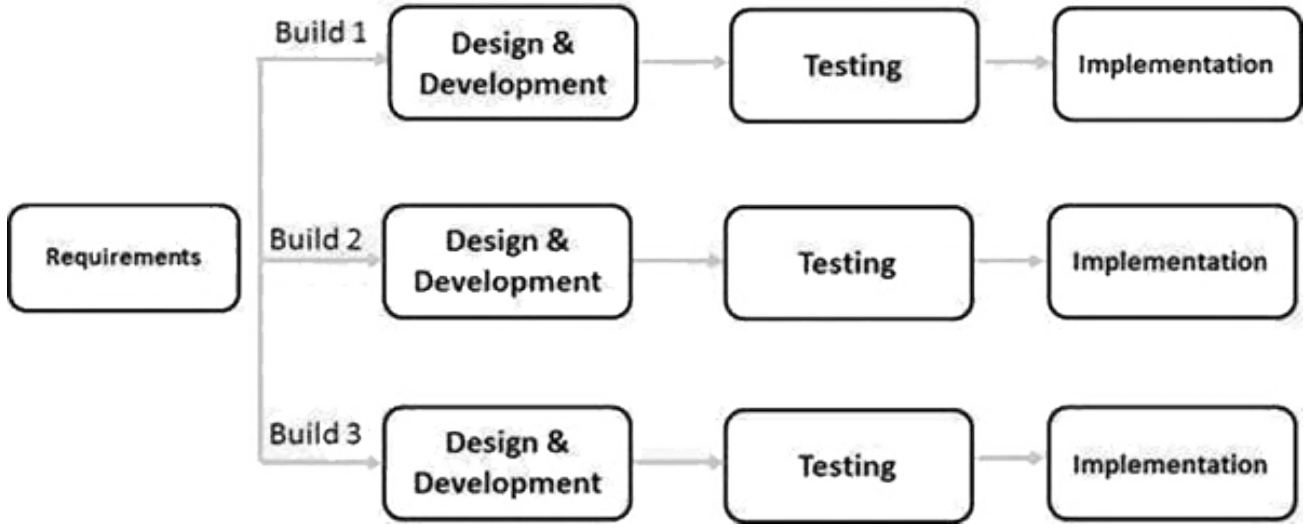
Project Development Model

- **Linear Model (Waterfall Model)**

- Process of software development is divided into separate phases.
- The outcome of one phase acts as the input for the next phase sequentially



Project Development Model

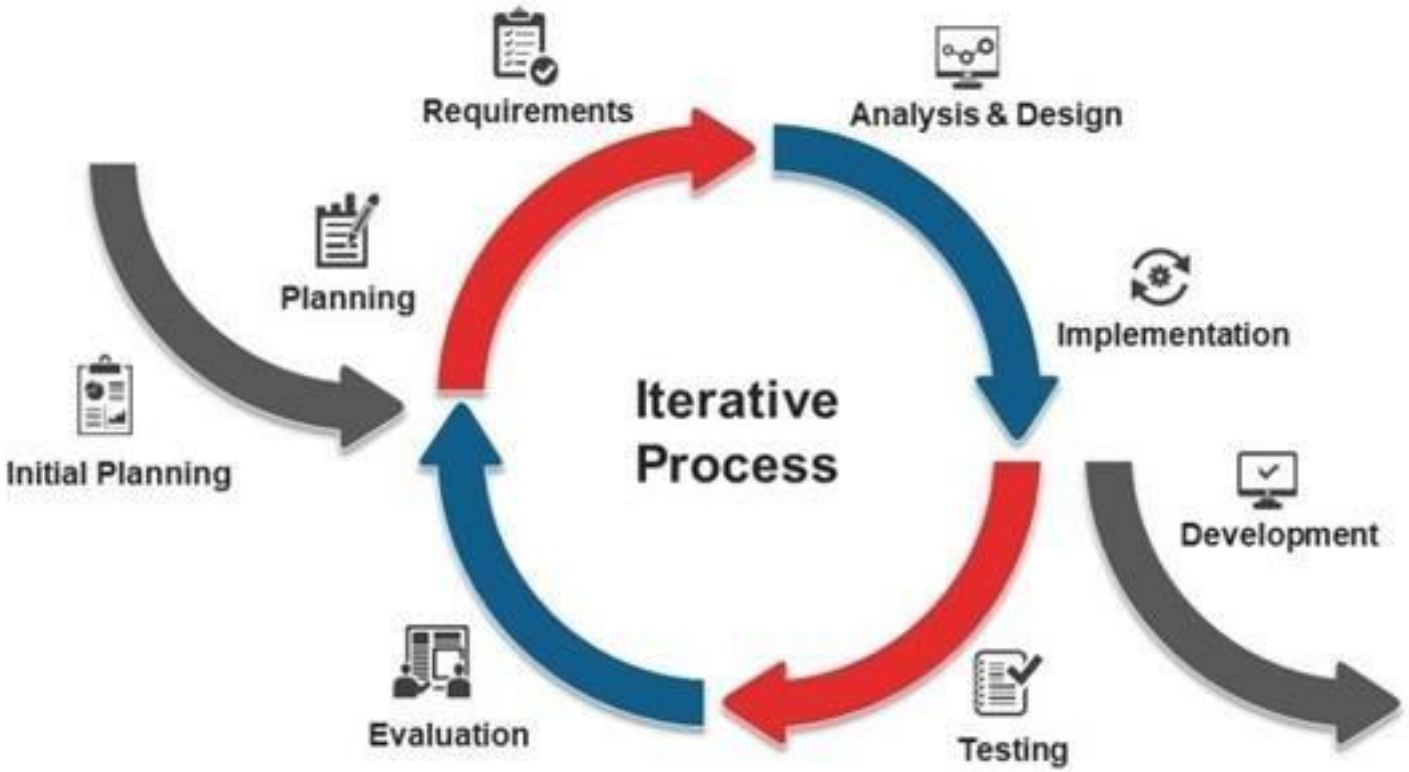


- **Incremental Model**

- It can be preferable to provide the software to the customer as a sequence of functional sub-systems over time rather than in one "big bang" at the conclusion of the project.
- It is **important** to have an outline plan in place for the entire system at the start.

Iterative(Agile) Process

- The **Iterative Process** is a software development approach where the system is built and improved **incrementally** through **repeated cycles (iterations)**. Each iteration results in a functional version of the product with enhancements based on feedback.



When to Use What?

Waterfall

Projects with :

- clear vision
- Fixed Requirements
- regulatory constraints

Incremental Model

Projects required:

- Requirements are mostly known but can evolve.
- Project can be divided into smaller, independent modules.
- Partial deployment is beneficial.

Iterative(Agile) Model

Projects required:

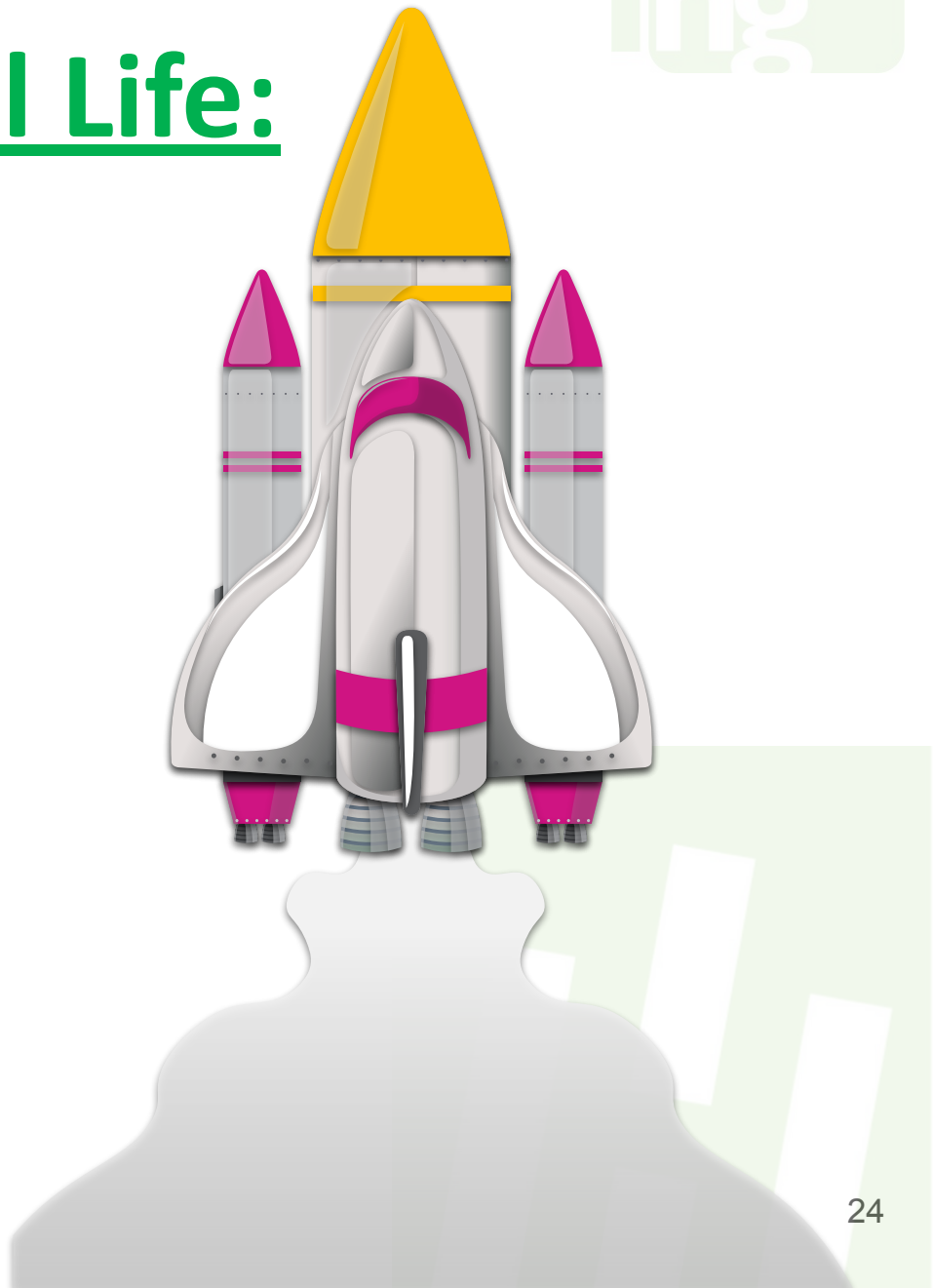
- Requirements are unclear or highly flexible.
- Rapid changes and continuous feedback are expected.
- Customer collaboration is necessary.

Waterfall Model Used in Real Life:

NASA's SPACE MISSIONS

Why WATERFALL?

- Space missions require strict planning
- clear requirements
- minimal changes after launch



Incremental Model Used in Real Life:

Mobile App Development:

Why Incremental?

Apps need:

- **continuous updates**
- **performance improvements**
- **A/B testing** to stay competitive

Examples:

- **Instagram**
- **Uber**
- **Spotify**
  - **Spotify uses Agile squads**—small, autonomous teams that work on different app features, improving the user experience iteratively

Case Study - Home Task

1. Banking System Upgrade for Secure Transactions

2. Campus Event Management App

THANK
YOU!

Project Simulation

Activities

Agenda

1. Select Topic & Method
   - Paper Boat
   - Paper Plane
   - Lego Building

2. Divide Group
   - Group A
   - Group B
   - Group C
   - Group D

3. Kick off Project
   - Start Project with selected topic

4. Compare the outcomes
   - Discuss which method work better!

Project: Paper House

Scenario:

Both teams are tasked with "building" a **3D paper house** within the **1.5-hour timeframe**. The house should have specific features (defined by the tutor), such as:

- A foundation
- Walls (at least 4)
- A roof
- Windows (at least 3)
- A door

Project: Paper House

Team A & C:
Waterfall Model (*Sequential Phase- No Iteration*)

Each Member works on a specific phase and hands it over to the next:

Roles	**Responsibilities**
Project Manager	Defines the house structure, timeline, and assigns tasks.
Designer	Creates a detailed house blueprint before any building starts.
Builder	Assembles the paper house model based on the finalized design.
Tester	Checks for structural stability, errors, and missing pieces.
Quality Analyst	Reviews the final model to ensure it meets initial design requirements.

👉 **Process:** No building starts until the design is finalized. Testing happens only after full assembly, meaning errors are costly to fix.

Project: Paper House

Team B & D:

Incremental Model

(*Built in Small Iterations & Improve over Time*)

Each Member works collaboratively in small iterations:

Roles	Responsibilities
Project Lead	Manages teamwork, sets goals for each iteration.
Designer/Builder1	Designs & builds the first small section (Iteration 1).
Designer/Builder2	Designs & builds the second section based on feedback (Iteration 2).
Tester	Continuously tests each completed section and provides feedback.
Quality Analyst	Ensures each iteration meets overall structure and stability requirements.

👉 **Process:** Small Lego parts are built and tested in phases, with improvements made in each iteration. Mistakes are easier to correct early.

Project: Paper House

Activity Steps:

- **Requirements Gathering & Planning (10 minutes):**
  - **Team Waterfall:** They carefully plan every aspect of the house construction. They create detailed blueprints with dimensions, materials, and the construction sequence. They also estimate the time needed for each stage.
  - **Team Incremental:** They start by planning the first increment. They identify essential features for the first version, such as the foundation and basic walls. They then plan the construction of this initial part.
- **Construction Phase 1 (15 minutes):**
  - **Team Waterfall:** They follow their detailed blueprint and start construction. They work in sequence, completing one stage before moving to the next. Communication focuses on following the plan. The house remains unseen until fully complete.
  - **Team Incremental:** They build the first part of the house, starting with the foundation and basic walls. They ensure it is functional and stable, even if unfinished. They can showcase their progress once this part is complete.
- **Review & Adaptation (10 minutes):**
  - **Team Waterfall:** They do not review yet. They continue building as planned.
  - **Team Incremental:** They review the first increment. They discuss successes, areas for improvement, and feedback. They plan the next increment, such as adding windows and a door.

Project: Paper House

Activity Steps:

- **Construction Phase 2 (15 minutes):**
  - **Team Waterfall:** They keep building as planned. If problems arise, they have limited flexibility to adapt.
  - **Team Incremental:** They build the second increment, using feedback from the review. They ensure the new addition is functional. They can showcase the added features.
- **Final Review & Presentation (30 minutes):**
  - **Each team** presents their completed paper house. They explain their process and discuss the challenges they faced and how they handled them.
- **Discussion & Comparison (15 minutes):**
  - The tutor leads a discussion comparing both approaches.
  - **Key points:**
    - **Visibility of Progress:** When did each team have a visible product?
    - **Adaptability to Change:** How easily could they adjust to new requirements?
    - **Risk Management:** Which team found problems later in the process?
    - **Collaboration:** How did their communication differ?

THANK
YOU!